

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

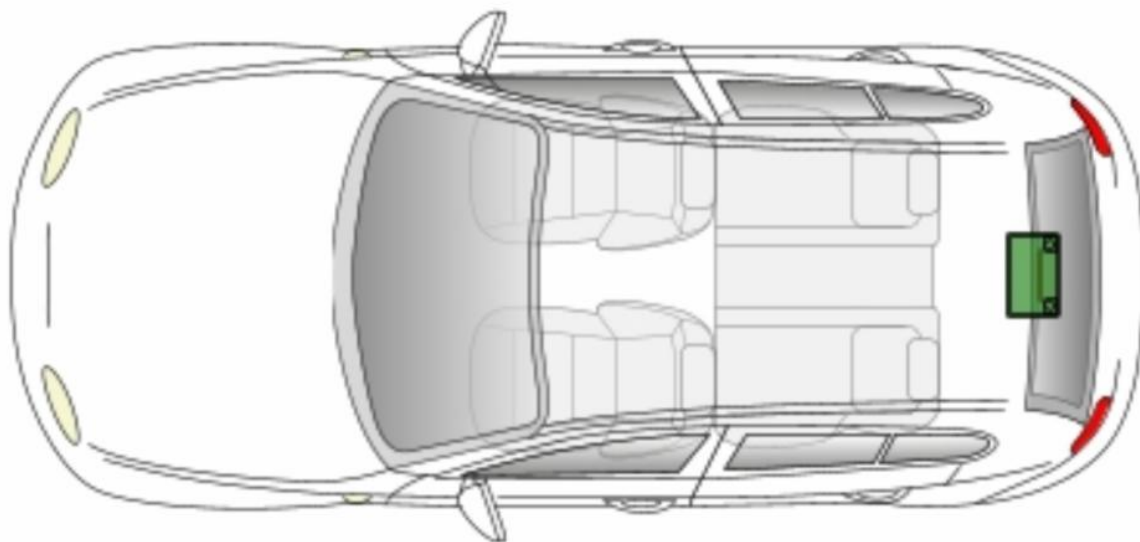
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

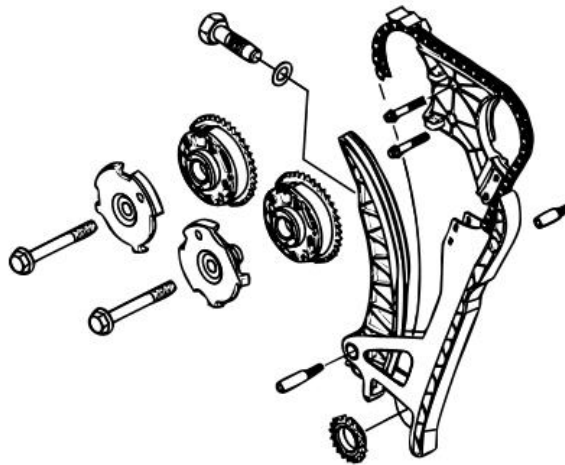
Battery location



Luggage compartment

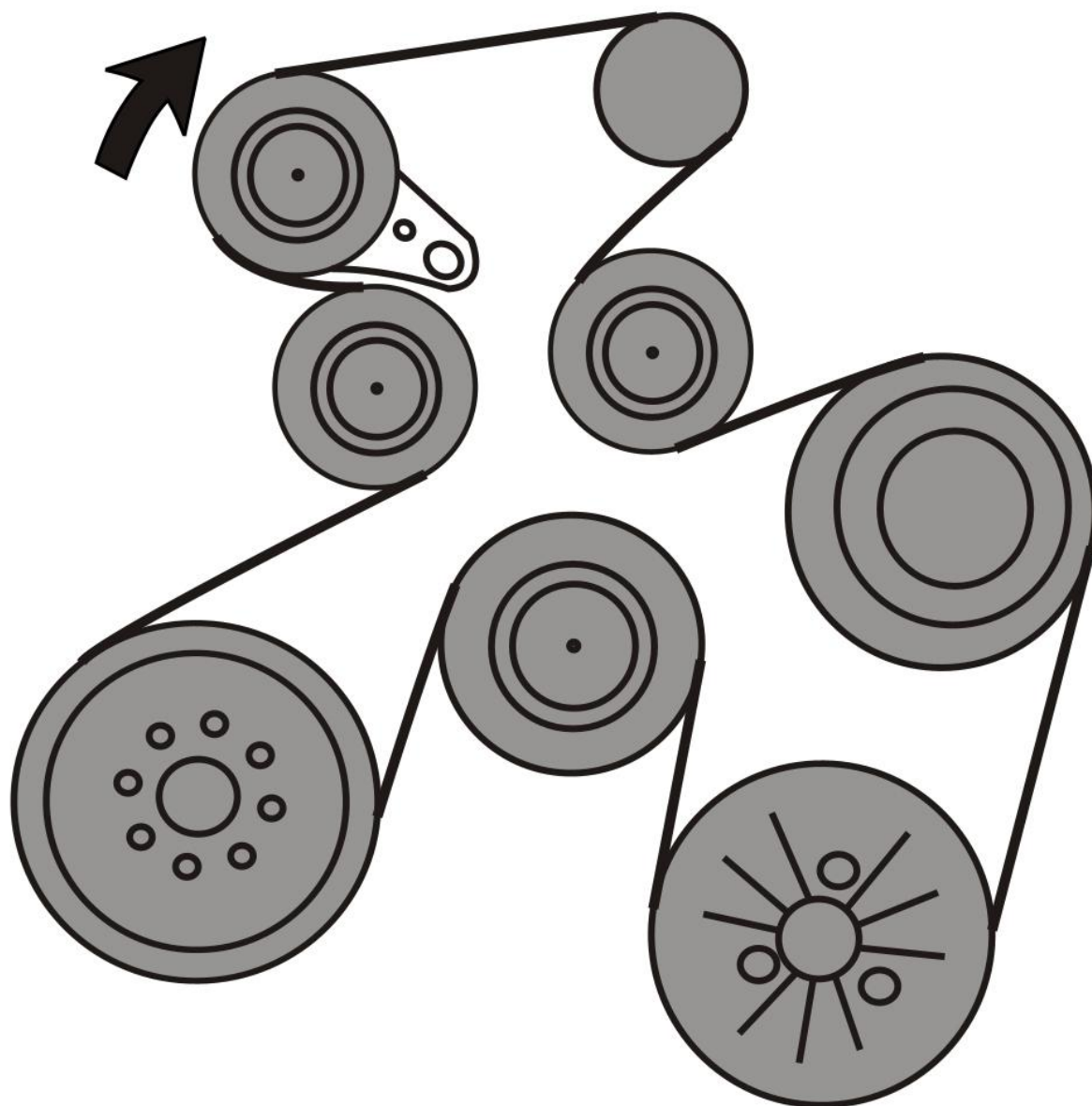
Timing chain layout

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Removal

- Disconnect the battery earth cable
- Remove the fan cowl
- Remove the engine upper cover
- Remove the ignition coils
- Disconnect the injection wiring
- Remove the injectors
- Remove the strut brace connecting the shock absorbers
- Remove the air duct
- Remove the cylinder head cover
- Remove the spark plugs
- Remove the chain tensioner piston
- Compress the chain tensioner piston twice to drain the oil
- Remove the engine lower cover
- Turn the tensioner pulley clockwise to release the tension
- Lock the tensioner; use a locking pin (11 3 340)
- Remove the ancillary drive belt
- Drive belt layout (With air conditioning)

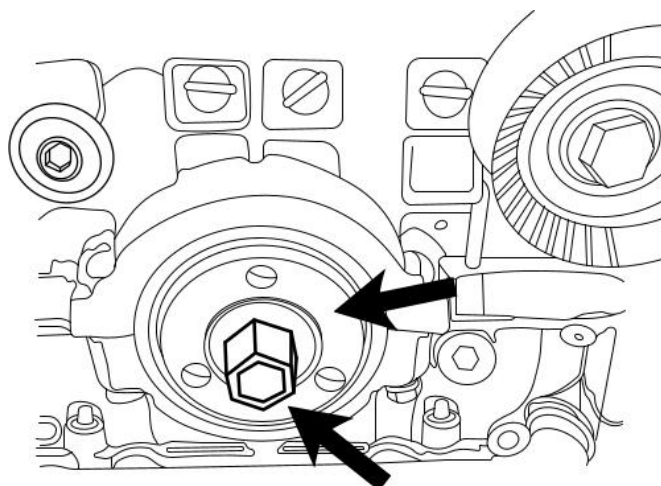
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Note: At removal, mark the direction of rotation; when refitting ensure the drive belt rotates in the same direction

Remove the crankshaft pulley

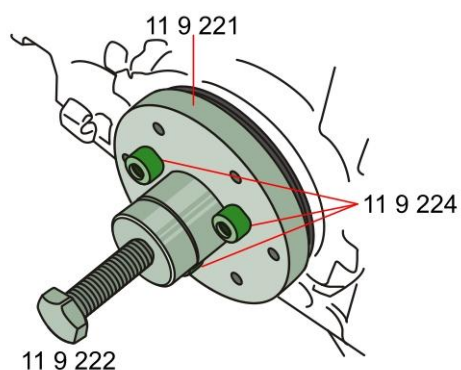
Note: Do not remove

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Fit the special tools (11 9 221, 11 9 222, 11 9 224)

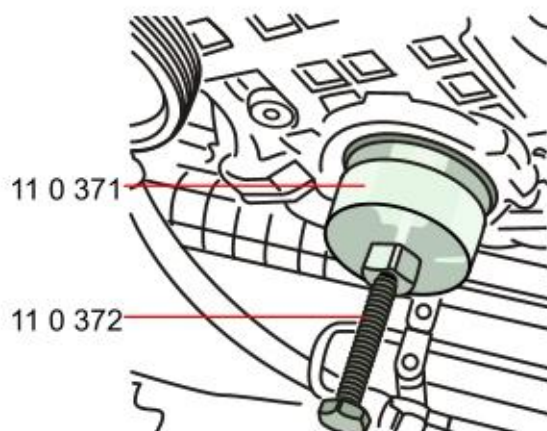
Tighten the special tool (11 9 224 - 20 Nm)



Remove the special tools (11 9 221, 11 9 222, 11 9 224)

Fit the special tools (11 0 371, 11 0 372)

Tighten the special tool (11 0 371 - 80 Nm)



Remove and discard the crankshaft oil seal

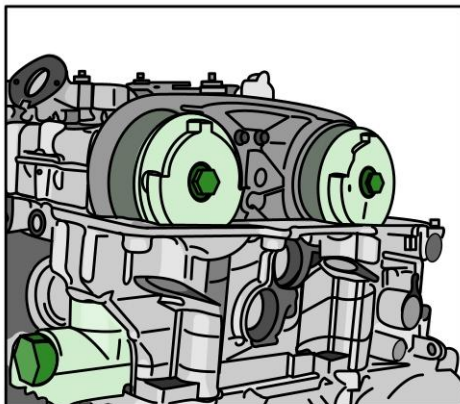
Remove the locking pin

Remove the ancillary drive belt tensioner (11 3 340)

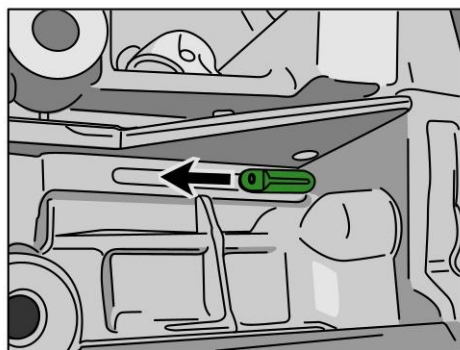
Remove the inlet camshaft timing adjuster

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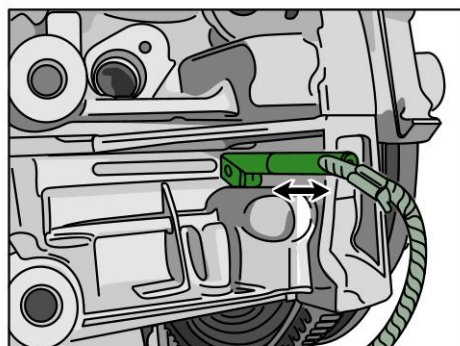
Remove the exhaust camshaft timing adjuster



Remove the access plug

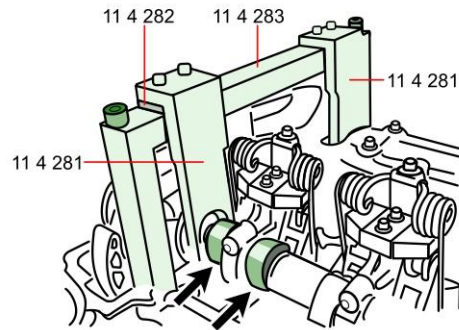


Fit the crankshaft TDC locking pin (11 0 300)



Fit the camshaft locking tools (11 4 281, 11 4 282, 11 4 283)

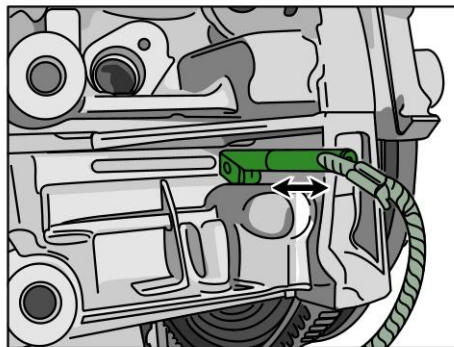
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Lock the flywheel

Use the special tools (11 8 660, 11 8 160)

Remove the crankshaft locking pin



Remove the crankshaft gearwheel centre bolt

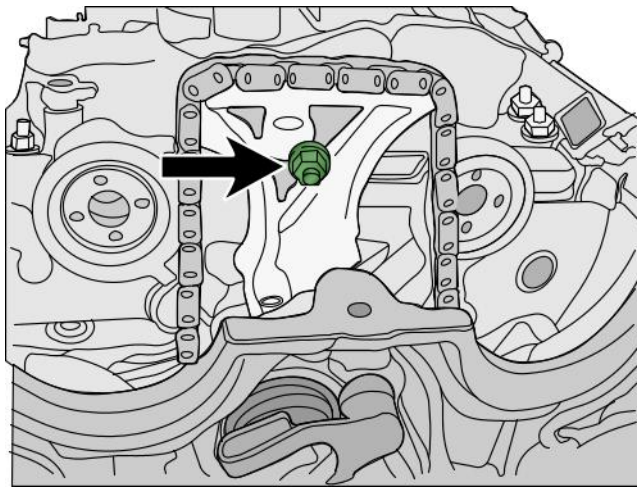
Remove the crankshaft hub

Bearing pins for guide and tensioner rails:

Remove the access plugs

Remove the guide pins

Remove the timing chain guide bolts from the cylinder head

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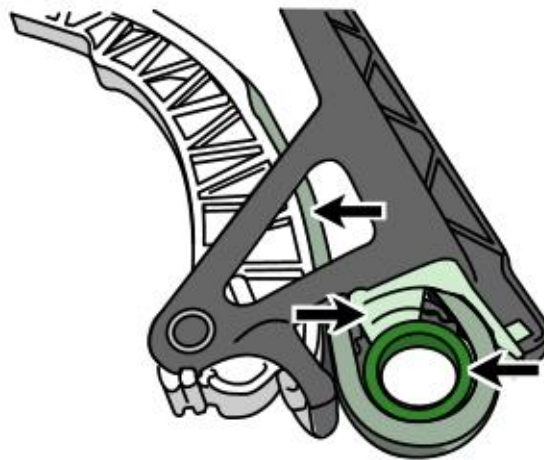
Remove the chain guide with the timing chain, guide rail and tensioner rail

Installation

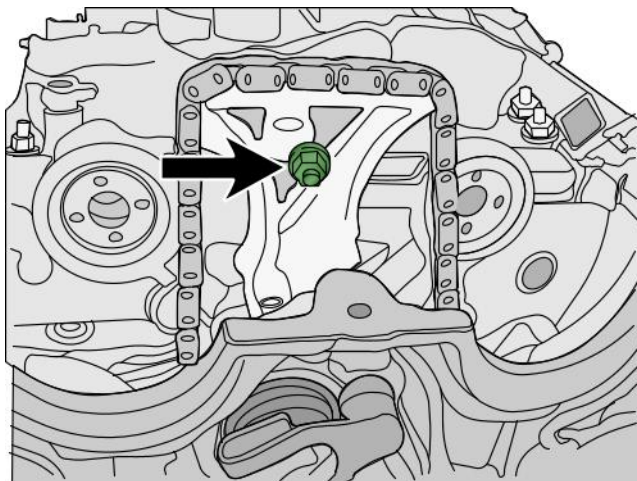
Crankshaft gearwheel:

Fit the shims

Refit the chain guide with the timing chain, guide rail and tensioner rail



Refit the timing chain guide bolts to the cylinder head (M6 x 35: 9 Nm)



Refit the crankshaft hub

Fit the special tool (00 9 140)

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Tighten the crankshaft gearwheel bolt (100 Nm + 360°)

Use new bolt(s)

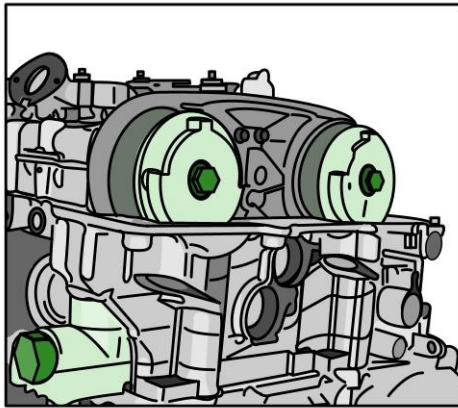
Bearing pins for guide and tensioner rails:

Refit the guide pins (M7: 14 Nm, M8: 20 Nm)

Refit the access plugs (M18 x 1.5: 25 Nm)

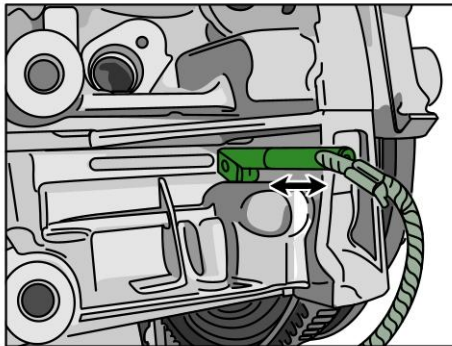
Refit the exhaust camshaft adjuster (AUS/EX)

Refit the inlet camshaft timing adjuster (EIN/IN)



Tighten the camshaft gearwheel bolts lightly until there is no play

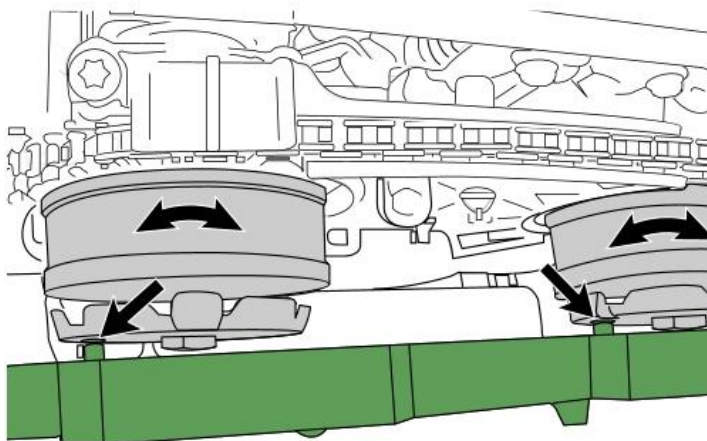
Fit the crankshaft TDC locking pin (11 0 300)



Fit the camshaft alignment tool (11 8 520)

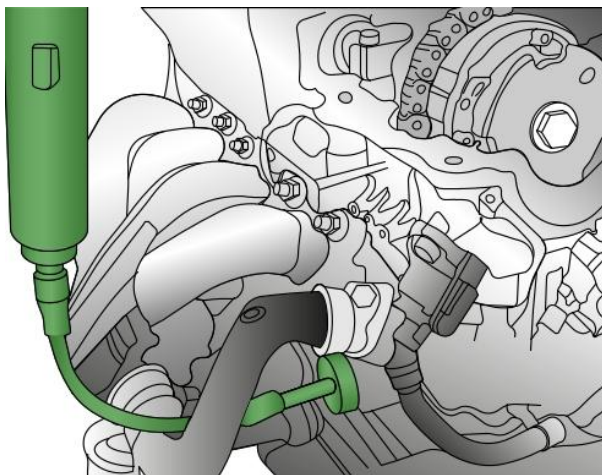
Align the camshaft gearwheel marks as shown

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Pre-tension the timing chain (0.6 Nm)

Use the special tools (00 9 250, 11 9 340)

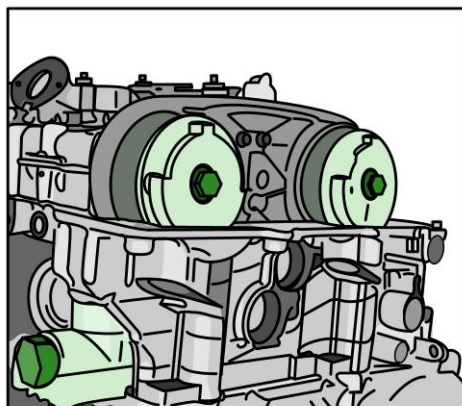


Tighten the camshaft gearwheel bolts (20 Nm + 180°)

Use new bolt(s)

Refit the chain tensioner piston (55 Nm)

Note: Fit a sealing ring



Remove all the locking tools

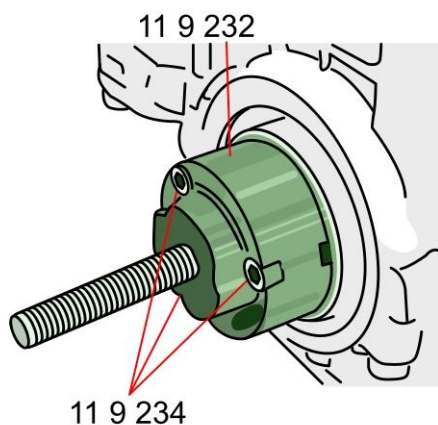
Turn the engine 2 rotations by hand

Check the timing again

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Refit the ancillary drive belt tensioner (M10 x 110: 40 Nm)

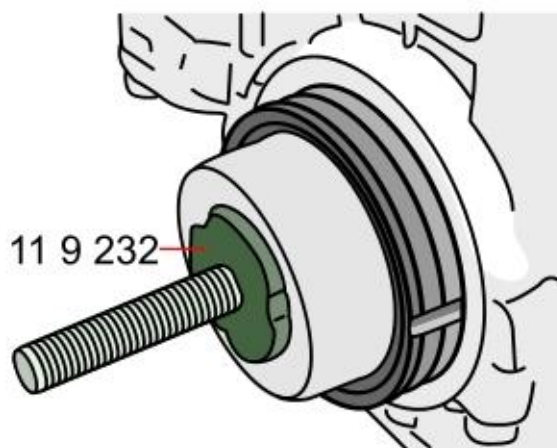
Fit the special tools (11 9 232, 11 9 234)



Apply grease

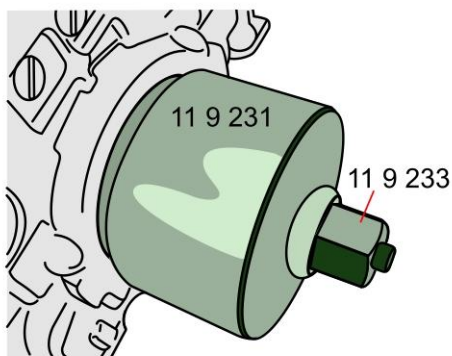
Fit the crankshaft seal over the special tool

Note: The seal should be aligned with the timing cover



Refit the crankshaft front radial seal

Use the special tool (11 9 231, 11 9 233)



Apply sealant

Refit the crankshaft pulley (25 Nm)

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- Turn the tensioner clockwise
- Lock the tensioner; use a locking pin (11 3 340)
- Fit the ancillary drive belt
- Remove the tensioner locking pin (11 3 340)
- Refit the engine lower cover
- Refit the spark plugs (M12 x 1.25: 23 Nm, M14 x 1.25: 30 Nm)
- Refit the cylinder head cover (9 Nm)
 - Use new bolt(s)
- Refit the air duct
- Refit the strut brace connecting the shock absorbers
- Refit the injectors
- Reconnect the injection wiring
- Refit and connect the ignition coils
- Refit the engine upper cover
- Refit the fan cowling
- Reconnect the battery earth cable

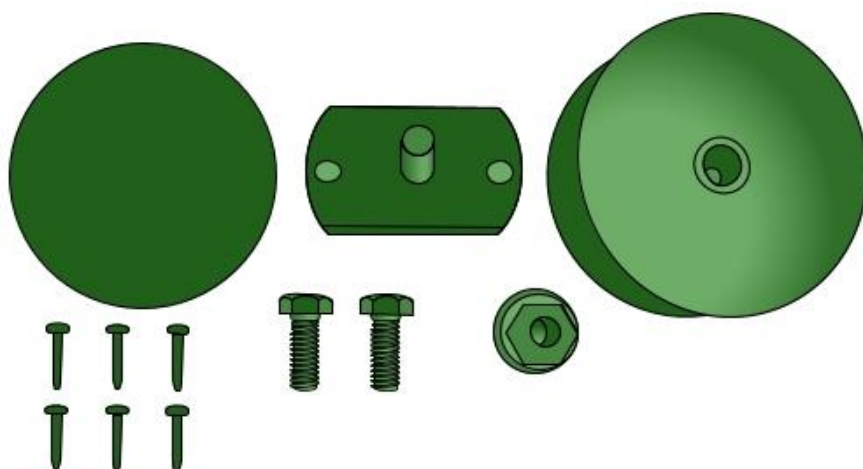
Torque settings

- Camshaft gearwheel
 - Camshaft gearwheel
 - Renew the bolts
 - Stage 1 20 (Nm)
 - Stage 2 180 (°)
- Crankshaft pulley
 - Crankshaft pulley (Outer bolts) 35 (Nm)
- Crankshaft gearwheel
 - Crankshaft gearwheel (Centre bolt)
 - Renew the bolts (Threads lightly oiled)
 - Stage 1 100 (Nm)
 - Stage 2 360 (°)
- Spark plugs
 - Spark plugs (M12) 20 - 26 (Nm)
 - Spark plugs (M14) 27 - 33 (Nm)
- Injector
 - Injector 11 - 15 (Nm)
- Fuel rail
 - Fuel rail 11 - 15 (Nm)
- Valve cover
 - Valve cover 8.5 (Nm)
- Ancillary drive belt tensioner pulley
- Timing chain tensioner
 - Timing chain tensioner (Upper)
 - Renew the seal
 - Stage 1 55 (Nm)
- Timing chain guide
 - Timing chain guide (Upper) 14 (Nm)
 - Timing chain guide (Lower) 20 (Nm)
- Chain guide: (M6 x 35: 9 Nm)
- Access plug: (M18 x 1.5: 25 Nm)

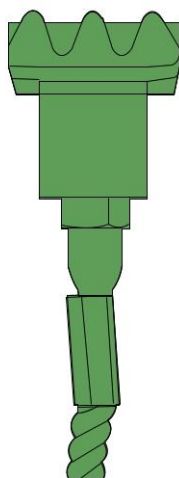
Special tools

- Extractor: (11 9 221, 11 9 222, 11 9 224, 11 0 371, 11 0 372)
- Crankshaft front radial seal fitting tool: (11 9 231, 11 9 232, 11 9 233, 11 9 234)

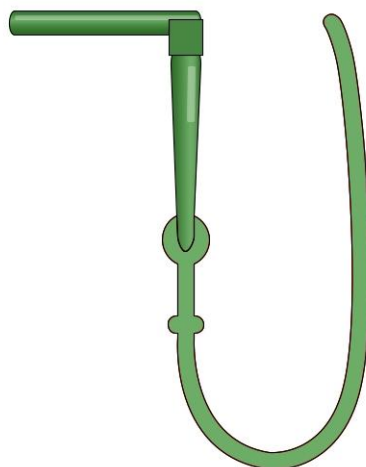
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Flywheel locking tool: (11 8 660, 11 8 160)

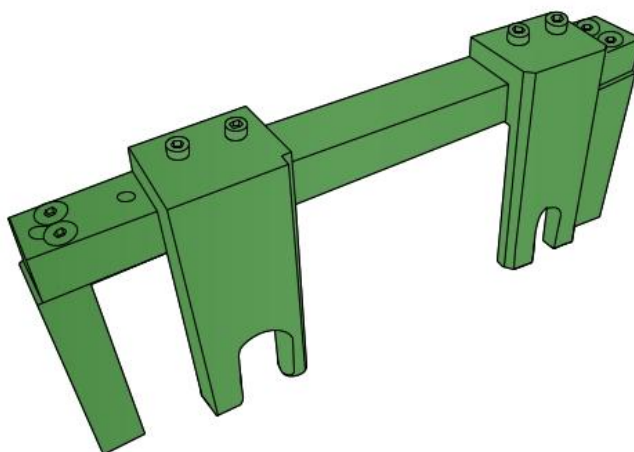


Crankshaft locking tool: (11 0 300)

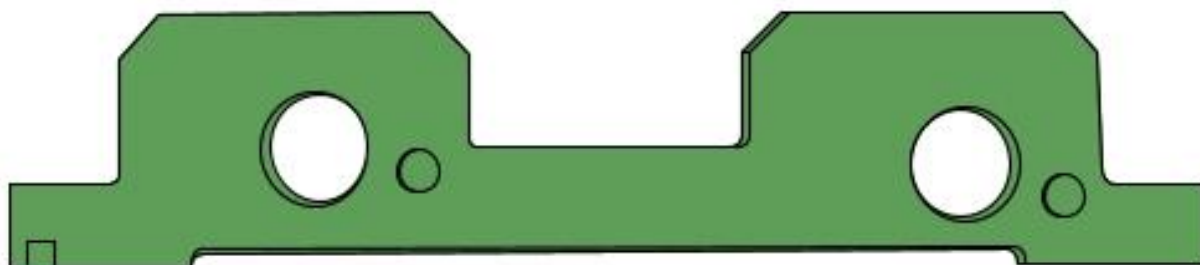


Camshaft locking tool: (11 4 281, 11 4 282, 11 4 283)

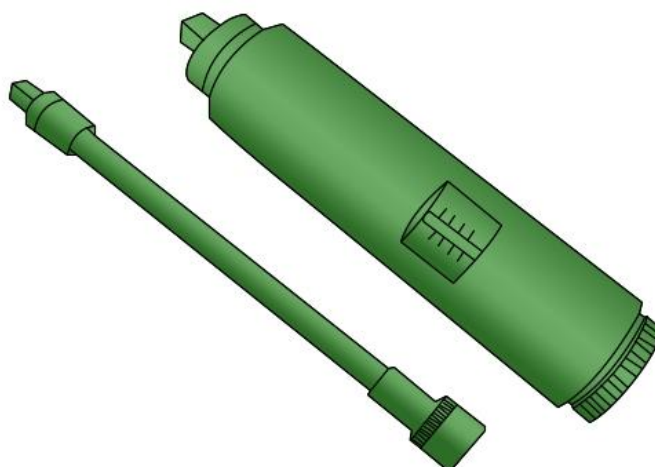
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Camshaft alignment tool: (11 8 520)

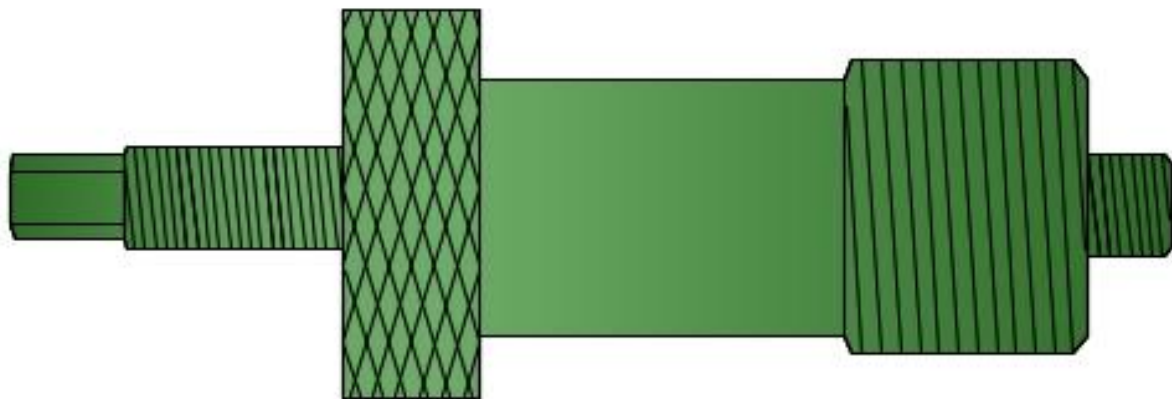


Torque wrench: (00 9 250)



Adaptor: (11 9 340)

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Tensioner locking pin: (11 3 340)

